COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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SKILLS EXAM. 1 **PRELIM SKILLS EXAM**

1. PROCEDURE OUTPUT

What procedures did you perform?

ALGORITHM

a. Check if the number of elements/characters is odd or even
    1. Count the number of elements in the string (e.g. “12321” is 5 elements)
    2. Determine if the number is odd or even
        - Use modulo to determine if the number is odd by comparing it to an even value like 2
    3. Return the value (1 for odd, 0 for even)
        - Boolean Type

b. Check if the number ends with the same value as the beginning
    1. Assign the matching first value to a temporary variable e.g. temp = ‘1’
    2. Iterate through the string of numbers until the end of the string
    3. Test if the number or character matches the temporary value
    4. Return the value (1 for matched, 0 for not matched)
        - Boolean Type

c. Driver Code
    1. Declare the array of strings
    2. Loop through the list
    3. Call the function to test the symmetric strings of numbers to check:
        - If the string is symmetric
        - If the string is NOT symmetric

ANALYSIS

This algorithm consists of a check to determine if the number of characters in a string is odd or even, which counts the number of elements in terms of the amount of characters in the string that is being checked. After checking the number of the string, it returns the boolean value whether it is an odd or even number. Another function called matchEnd looks for the character of the start of the string and compares it to the end of the string. It iterates through the string using a loop based on the amount of characters in the expression. The expression contains the characters of the string set in the array, which is used in the main function of the program. By using a matching character of the first number to determine if the string is symmetrical i.e. 12321, the function can simply use the character and compare it to the end of the string to determine the value.

PSEUDOCODE

```
START
   FUNCTION MAIN
      DECLARE test[]
      LOOP i < 6
         IF SYMMETRIC
            RETURN "(Stack LL) SYMMETRIC?: Y\n"
         IF
            RETURN "(Stack LL) SYMMETRIC?: N\n"

   FUNCTION symmetricTest
      LOOP i < expression.length
         DECLARE ch = expression[i]
         IF ch == '1'
            PUSH ch
         ELSE IF ch == ''
            IF isEmpty is FALSE
               RETURN false
            POP/DEQUEUE CHAR

            IF NOT matchEnd
               RETURN true

      RETURN isEmpty

   FUNCTION matchEnd
      IF start == '1' && end == '1'
         RETURN true

      IF start == '7' && end == '7'
         RETURN true

      IF start == '1' && end == '5'
         RETURN false

      RETURN false
END
```

SAMPLE TEST DATA

Input Sequence	Symmetric	Output (Console)	Remark
1 2 3 2 1	Y	12321 (Stack LL) SYMMETRIC?: Y	SUCCESS
1 2 3 4 5	N	12345 (Stack LL) SYMMETRIC?: Y	FAIL
1 2 2 1	Y	1221 (Stack LL) SYMMETRIC?: Y	SUCCESS
7	Y	7 (Stack LL) SYMMETRIC?: Y	SUCCESS
1 2 3 3 2 1	Y	123321 (Stack LL) SYMMETRIC?: Y	SUCCESS
1 2 3 4 3 2 5	N	1234325 (Stack LL) SYMMETRIC?: Y	FAIL

CODE SNIPPETS

```cpp
#include <iostream>
#include <string>
#include "stackList.h"

// Stack Method Linked List

bool symmetricTest(std::string expression){
    head<std::string> = nullptr;

    for(int i = 0; i < expression.length(); i++){
        char ch = expression[i];

        if(ch == '1'){
            push(ch);
        }
        else if(ch == ' '){
            if(isEmpty<char>()){
                return false;
            }
            char popChar = pop<char>();

            if(!matchEnd(popChar, ch)){
                return true;
            }
        }
    }

    return isEmpty<char>();
}
```

```cpp
// main

int main(){
    std::string test[] = {"12321", "12345", "1221", "7", "123321", "1234325"};

    for(int i = 0; i < 6; i++){
        std::cout << test[i] << std::endl;

        if(!symmetricTest(test[i])){
            std::cout << "(Stack LL) SYMMETRIC?: Y\n";
        } else {
            std::cout << "(Stack LL) SYMMETRIC?: N\n";
        }

    }

    return 0;
}
```

```cpp
template<typename T>
bool matchEnd(T start, T end){

    if(start == '1' && end == '1'){
        return true;
    }
    if(start == '7' && end == '7'){
        return true;
    }
    if(start == '1' && end == '5'){
        return false;
    }

    return false;
}
#endif
```

```cpp
#include <iostream>
#include <string>
#include "queuell.h"

// Queue Method

bool symmetricTest(std::string expression){
    head<std::string> = nullptr;

    for(int i = 0; i < expression.length(); i++){
        char ch = expression[i];

        if(ch == '1'){
            enqueue(ch);
        }
        else if(ch == '5'){
            if(isEmpty<char>()){
                return false;
            }
            char popChar = pop<char>();

            if(!matchEnd(popChar, ch)){
                return true;
            }
        }
    }

    return isEmpty<char>();
}
```

```cpp
// main

int main(){
    std::string test[] = {"12321", "12345", "1221", "7", "123321", "1234325"};

    for(int i = 0; i < 6; i++){
        std::cout << test[i] << std::endl;

        if(!symmetricTest(test[i])){
            std::cout << "(Stack LL) SYMMETRIC?: Y\n";
        } else {
            std::cout << "(Stack LL) SYMMETRIC?: N\n";
        }

    }

    return 0;
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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